

Problem 1: Breast Cancer Classification

Dataset: <https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>

Objective

Predict whether a tumor is:

- Malignant (M)
- Benign (B)

Tasks

1. Load the dataset.
2. Drop unnecessary columns (e.g., id, Unnamed: 32).
3. Convert target labels (M $\rightarrow$ 1, B $\rightarrow$ 0).
4. Split the dataset (80/20).
5. Standardize the features.
6. Train an SVM with a **Linear Kernel**.
7. Evaluate using:
 - Accuracy
 - Precision
 - Recall

Problem 2: Mushroom Classification

Dataset: <https://www.kaggle.com/datasets/uciml/mushroom-classification>

Objective

Predict whether a mushroom is:

- Edible
- Poisonous

Tasks

- Handle categorical variables.
- Encode all features.
- Train Linear SVM.
- Evaluate performance.
- Compare with Decision Tree.

Problem 3: Bank Marketing

Dataset: <https://www.kaggle.com/datasets/janiobachmann/bank-marketing-dataset>

Objective

Predict whether a customer will subscribe to a term deposit.

Tasks

- Handle categorical variables.
- Feature scaling.
- Train Linear SVM.
- Print evaluation metrics.

Problem 4: Heart Disease Prediction

Dataset: <https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>

Objective

Predict whether a patient has heart disease.

Tasks

Train four SVM models:

- Linear
- Polynomial
- RBF
- Sigmoid

Compare

- Accuracy
- Precision
- Recall

Problem 5: Customer Churn Prediction

Dataset: <https://www.kaggle.com/datasets/blatchar/telco-customer-churn>

Objective

Predict whether a customer will churn.

Tasks

- Handle missing values.
- Encode categorical columns.
- Scale numerical columns.
- Train SVM.
- Tune hyperparameters.

Problem 6: Credit Card Fraud Detection

Dataset: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

Objective

Detect fraudulent transactions.

Tasks

- Handle imbalanced data. (there is a parameter in SVC to address this)
- Train SVM.
- Use:
 - Precision
 - Recall
 - F1-score